

o-Phosphate–P in Seawater

1 SCOPE

This method covers the determination of o-Phosphate–P in seawater.

2 RANGE OF APPLICATION

Range	0.1 – 7.0 μM (3.1 – 217 $\mu\text{g P/L}$)
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3 METHOD DETECTION LIMIT

By EPA Procedure	MDL = 0.015 μM (0.464 $\mu\text{g P/L}$)
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4 METHOD PRINCIPLE

Reaction with acidic molybdate in the presence of antimony forms an antimony phospho-molybdate complex, which is then reduced by ascorbic acid to phosphomolybdenum blue and is measured photometrically at 880 nm.

5 REFERENCES

Methods for the Determination of Inorganic Substances in Environmental Samples, USEPA 600/R 93/100, August 1993: Method 365.1, Rev. 2.0.

Standard Methods for the Examination of Water and Wastewater, APHA/AWWA/WEF, method 4500-P-F (2012 forward).

6 INTERFERENCES

Interferences can come from a variety of analytes including barium, lead, iron, silver, silica, nitrite, sulfide, arsenate, hexavalent chromium, and arsenic; however the effects of these are usually negligible in natural waters. See Section 11.1, Chemistry Troubleshooting Tips for specifics.

Salt errors for samples ranging from 5-20% salt content were found to be less than 1%.

Arsenate is determined similarly to phosphorus and should be considered when present in concentrations higher than phosphorus. However, at concentrations found in sea water, it does not interfere.

Filtration of samples for total or total hydrolysable phosphorus should only occur after digestion.

Bias from sample background color or turbidity is correctable, using the sample blanking feature of AQ software.

7 REAGENTS: CHEMICALS REQUIRED AND SAFETY INFORMATION

Consult the material safety data sheets (SDS) for details on safe handling of chemicals.

<u>Chemical</u>	<u>Formula</u>	<u>C.A.S. No.</u>	<u>Safety Classification</u>
Ammonium molybdate tetrahydrate	$(\text{NH}_4)_6\text{Mo}_7\text{O}_{24} \cdot 4\text{H}_2\text{O}$	12054-85-2	--
Ascorbic acid	$\text{C}_6\text{H}_8\text{O}_6$	50-81-7	--
EDTA disodium salt dihydrate*	$\text{C}_{10}\text{H}_{14}\text{N}_2\text{Na}_2\text{O}_8 \cdot 2\text{H}_2\text{O}$	6381-92-6	GHS07, GHS09
Potassium antimonyl tartrate trihydrate	$\text{C}_8\text{H}_4\text{K}_2\text{O}_{12}\text{Sb}_2 \cdot 3\text{H}_2\text{O}$	28300-74-5	GHS07, GHS09
Potassium phosphate monobasic	KH_2PO_4	7778-77-0	--
Sodium bicarbonate*	NaHCO_3	144-55-8	--
Sodium chloride*	NaCl	7647-14-5	--
Sodium hydroxide*	NaOH	1310-73-2	GHS05
Sulfuric acid, concentrated	H_2SO_4	7664-93-9	GHS05

GHS01: Danger – Explosive

GHS02: Danger – Flammable

GHS03: Danger – Oxidizing

GHS04: Warning – Compressed Gas

GHS05: Warning/Danger – Corrosive

GHS06: Danger – Toxic

GHS07: Warning – Irritant

GHS08: Danger – Health Hazard

GHS09: Warning/Danger – Aquatic Toxicity

Note: Persons using this method should be familiar with normal laboratory practice. This method does not purport to address all of the safety problems, if any, associated with its use. It is the responsibility of the user to establish appropriate safety and health practices and to ensure compliance with local regulatory conditions.

* Only required for artificial seawater preparation

* Used only for Special Cleaning Solution

8 REAGENT SOLUTIONS

Use reagent-grade chemicals, certified for analytical or general laboratory use. SEAL recommends using ACS Reagent Grade whenever possible. Reagents of technical or commodity grade must be validated by the user. Use high-purity reagent water, distilled or deionized, and free from organic contamination, therefore Grade 1 (ISO Standard 3696) or better than Type II (ASTM Standard D1193) waters are suitable.

8.1 SULFURIC ACID, 5 normal

Sulfuric acid, concentrated	70 mL
DI water	dilute to 500 mL

Slowly add 70 mL concentrated sulfuric acid to about 400 mL of deionized water. **CAUTION** the container will become very warm. Cool to room temperature and dilute to 500 mL with deionized water.

Store at room temperature. This solution is available commercially. Stable for 6 months.

8.2 POTASSIUM ANTIMONYL TARTRATE, 1.37 g/L

Potassium antimonyl tartrate trihydrate	0.137 g
DI water	dilute to 100 mL

Dissolve 0.137 g potassium antimonyl tartrate trihydrate in about 80 mL deionized water. Dilute to 100 mL and mix well.

Store at 4°C in a plastic bottle. Stable for 1 month.

8.3 AMMONIUM MOLYBDATE (3% w/v)

Ammonium molybdate tetrahydrate	3.0 g
DI water	dilute to 100 mL

Dissolve 3.0 g ammonium molybdate tetrahydrate in approximately 80 mL deionized water. Dilute to 100 mL.

Store at 4°C in a plastic bottle. Stable for 1 month. Discard this reagent if it becomes discolored or turbid.

8.4 WORKING PHOSPHATE COLOR REAGENT

Sulfuric acid, 5 normal (above)	50 mL
Ascorbic acid, 15 g/L (above)	20 mL
Potassium antimonyl tartrate, 1.37 g/L (above)	10 mL
Ammonium molybdate, 3% w/v (above)	20 mL

To a plastic beaker or bottle, add 50 mL sulfuric acid, 5 normal followed by 20 mL of ascorbic acid, 15 g/L. Swirl to mix. Then add 10 mL potassium antimonyl tartrate, 1.37 g/L and swirl to mix. Then add 20 mL ammonium molybdate, 3% (w/v). Swirl the contents. This solution will be slightly yellow.

NOTE: Molybdate must be added last.

Make fresh daily and keep refrigerated when not in use. Discard if the reagent turns blue or becomes turbid.

8.5 ASCORBIC ACID, 15 g/L

Ascorbic acid, fine granular	1.5 g
DI water	dilute to 100 mL

Dissolve 1.5 g of ascorbic acid, fine granular, in about 80 mL deionized water. Dilute to 100 mL and mix well.

Store at 4°C in a plastic bottle when not in use. Prepare fresh daily.

8.6 SPECIAL CLEANING SOLUTION

Sodium hydroxide	4.0g
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EDTA disodium salt dihydrate	1.0 g
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DI Water	100 mL
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Dissolve 4.0 g of sodium hydroxide in approximately 80 mL of water. Add 1.0 g of EDTA disodium salt dihydrate. Dilute to 100 mL and mix well.

Store at room temperature. Stable for 2 months.

9 STANDARD SOLUTIONS

9.1 PHOSPHATE STOCK STANDARD SOLUTION (5000 μM / 154.5 mg P/L)

Potassium phosphate monobasic 0.6804 g
DI water dilute to 1 L

Dissolve 0.6804 g potassium phosphate monobasic, anhydrous (dried at 105°C) in about 800 mL deionized water in a 1 Liter volumetric flask. Up to 10 mL sulfuric acid, 5 normal, may be added for preservation. Dilute to the mark with deionized water.

Store at 4°C in a plastic container. Stable for 1 month.

9.2 CALIBRATION DILUENT (SYNTHETIC SEAWATER)

Sodium chloride 35 g
Sodium hydrogen carbonate 0.2 g
DI water dilute to 1 L

Dissolve 35 g of sodium chloride and 0.2 g of sodium hydrogen carbonate in about 800 mL of deionized water in a 1 liter plastic volumetric flask. Dilute to 1 liter with deionized water.

***Note:** The concentration of sodium chloride can be adjusted to match the salinity of the seawater samples.

Store at room temperature in a plastic container. Stable for 1 month.

9.3 PHOSPHATE INTERMEDIATE STANDARD SOLUTION (250 μM / 7.725 mg P/L)

PHOSPHATE Stock Standard (5000 μM P, above) 10 mL
Calibration diluent (above) dilute to 200 mL

Pipet 10 mL PHOSPHATE–P Stock Standard (5000 μM P) to a 200 mL volumetric flask. Fill to the mark with calibration diluent and invert to mix.

Refrigerate and store in a plastic container. Prepare fresh monthly.

9.4 AUTO CALIBRATION STANDARD – Phosphate-P (7 μM / 216.3 μg P/L)

PHOSPHATE Intermediate Standard (250 μg P, above) 7.0 mL
Calibration diluent (above) dilute to 250 mL

Transfer 7.0 mL PHOSPHATE–P Intermediate Standard (250 μg P) to a 250 mL volumetric flask. Fill to the mark with calibration diluent and invert to mix.

Store at 4°C in a plastic container. Stable for 1 week.

9.5 MANUAL CALIBRATION

These are suggested concentrations for manual calibration standards.

Concentration, μM P/L	0	0.14	0.35	0.7	1.75	3.5	5.25	7.0
Volume (mL) Auto Calibration Standard Solution diluted as indicated with deionized water	--	2.0	5.0	10	25	50	75	100
Final Volume, mL	100	100	100	100	100	100	100	100

Prepare weekly.

10 SAMPLE PRESERVATION AND STORAGE

Samples may be collected into plastic containers.

Samples to be reported for ortho-phosphate for EPA's drinking water compliance monitoring programs should NOT be filtered, digested, or undergo hydrolysis. The samples should be cooled to $\leq 6^{\circ}\text{C}$ and must be analyzed within 48 hours of collection. Do not use acid for preservation.

Samples to be reported for dissolved ortho-phosphate for EPA's wastewater compliance monitoring programs must be filtered immediately, then cooled to $\leq 6^{\circ}\text{C}$ and must be analyzed within 48 hours of collection. Do not use acid for preservation.

Samples to be reported for total-phosphorus for EPA's wastewater compliance monitoring programs must be preserved with sulfuric acid to a $\text{pH} < 2$, cooled to $\leq 6^{\circ}\text{C}$ and must be analyzed within 28 days. Digestion is required as described in EPA method 365.1, 365.2 or 365.3. Do not filter prior to digestion. After digestion, neutralize with sodium hydroxide as described in EPA method 365.1 or 365.2. If pH neutralization is not desired, analyze the finished digests using SEAL method EPA-119-C instead. Call SEAL Analytical Technical Support for assistance.

11 PROCEDURE

Prepare standards and reagents as described in Sections 8 and 9. The laboratory must determine standard concentrations that are most suitable to their needs for calibration within the range of application for this method. The concentrations listed within this method are only suggestions.

Any reagents having visible particulates should be filtered.

Pour calibration standards, quality control solutions, and samples into sample cups and reagent vessels, according to the user-defined work list.

Before running the analysis batch, check the analyzer for sufficient reagent water, fresh reaction segments, correct location of reagent wedges and a waste collection bottle in place.

After o-phosphate analyses are finished for the day, clean the cuvette to remove deposits. Fill a reagent wedge with SEAL Cuvette Cleaning Solution (p/n 5676) and place it in Position 1 of the reagent rack. Within SEAL discrete analyzer software, go to Maintenance—Cuvette Functions and select Extra Wash.

11.1 CHEMISTRY TROUBLESHOOTING TIP

Glassware should be washed with 1:1 HCl, rinsed with phosphate color reagent, and rinsed thoroughly with deionized water. If glassware is reserved solely for the determination of phosphorus, then treatment with 1:1 HCl and phosphate color reagent is only required occasionally. Commercial detergent should never be used.

As much as 50 mg Fe^{3+}/L , 10 mg Cu^{2+}/L , and 10 mg SiO_2/L can be tolerated. High silica concentrations cause positive interference and high iron concentrations can cause precipitation of, which causes a subsequent loss of phosphorus. For silica concentrations of 20, 50, and 100 mg/L, phosphorous results will be elevated by 0.0005, 0.015, and 0.025 mg/L, respectively.

Eliminate interferences from nitrite and sulfide by adding an excess of bromine water or a saturated potassium permanganate solution.

Hexavalent chromium and nitrite interfere to give results about 3% low at concentrations of 1 mg/L and 10 to 15% low at 10 mg/L. Sulfide (Na_2S) and silicate do not interfere at concentrations of 1.0 and 10 mg/L, respectively.

The molybdate to acid ratio is very important for the phosphate chemistry. If this ratio is off, then the curve will be quadratic. If curve is quadratic, check age of the acid and molybdate solutions. Remake color reagent and stocks.

If running a lot of phosphate on the AQ400, it is recommended to run the Special Cleaning Solution (section 8.6) to clean the cuvette. To do this, fill a reagent wedge with Special Cleaning Solution and place it in Position 1 of the reagent rack. Within SEAL AQ software, go to Maintenance—Cuvette Functions and select Extra Wash.

For best results, pour ortho-phosphate color reagent fresh before each run.

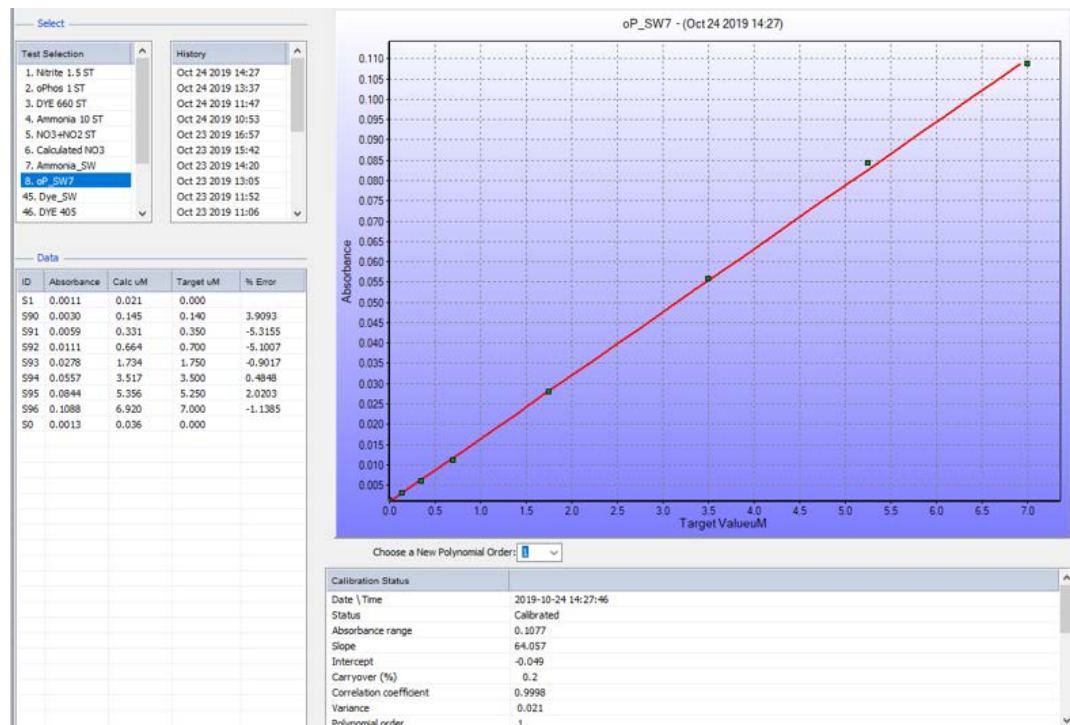
12 TEST PARAMETERS

PARAMETER	AQ SETTING
Test name	o-PHOS_SW
Units	uM
Decimals	3
Test type	End Point
Sample volume (μL)	500
Water volume (μL)	0
Number of mixes	2
Cuvette primes	4
Cuvette washes	2
Baseline on Wash	Ticked
Reaction time (seconds)	540
Wavelength (nm)	880
Polynomial order	1
Number of reagents	1
1. PO4 color (μL)	40
Advanced Test Parameters	☒ Extra Debubbling Action

DEFINITIONS:

Test name:	Name which appears on the final report
Units:	Concentration units appearing against the reported result
Decimals:	Number of decimal places in the reported result
Test type:	The type of test the software will perform
Sample volume:	Volume (in microliters) of the sample used in the test
Water volume:	Volume (in microliters) of water from the water reservoir used in the test
Number of mixes:	Number of mixes of the sample + reagent in the reaction well
Cuvette primes:	Number of times the cuvette is flushed with reagent (no sample) prior to sampling
Cuvette washes:	Number of washes of the cuvette between aspirations of finished reaction liquid
Baseline on Wash:	Water baseline measured between aspirations to provide drift correction
Reaction time:	Time (in seconds) between addition of sample + reagent and measurement of the reaction mixture
Wavelength:	Interference filter wavelength to be used in the test
Polynomial order:	Polynomial order to be used when fitting the standard points to the standard graph (1 = linear)
Number of reagents:	Number of reagents defined for the test
Reagent volume:	Volume (in microliters) of the reagent used in the test

13 STANDARDIZATION



Max. Blank Absorbance (S1): 0.005 AU

Sensitivity Range (7 μ M P): 0.087 – 0.13 AU

DEFINITIONS:

Abs. Range: Difference between the absorbance of the highest standard and the absorbance of the blank

Variance: Calculated as follows:

$$\text{Variance} = \frac{\sum(\text{Deviations, i.e., errors})^2}{\sum(\text{Observed values})^2} \times 100$$

Carryover: Calculated as follows:

$$\text{Carryover (\%)} = \frac{(A-B)}{C} \times 100\%$$

Where, A = absorbance of carryover blank (S0)

B = absorbance of blank (S1)

C = calibration absorbance range

14 PERFORMANCE VALIDATION DATA

14.1 DETECTION LIMIT STUDY

Replicates of 0.1 μM (3.1 $\mu\text{g/L}$ P/L) o-phosphate standard in synthetic seawater***

Concentration (mg P/L)				
	0.090	0.098	0.109	0.104
	0.078	0.099	0.117	0.110
	0.083	0.107	0.108	0.109
	0.086	0.095	0.110	0.105
	0.086	0.099	0.106	0.106
	0.080	0.103	0.112	0.112
	0.079	0.099	0.109	0.114
	0.082	0.090	0.104	0.110
	0.089	0.096	0.111	0.126
	0.081	0.097	0.117	0.114
	0.092	0.108	0.120	0.110
	0.083	0.094	0.109	0.112
	0.090	0.094	0.108	0.114
	0.087	0.096	0.110	0.113
	0.086	0.094	0.119	0.108
Statistics				
Mean	0.085	0.098	0.111	0.111
Std. Deviation	0.004	0.005	0.005	0.005
MDL* (μM)	0.011	0.013	0.013	0.014
MDL* ($\mu\text{g P/L}$)	0.352	0.399	0.389	0.424

14.2 PERFORMANCE STUDY

Replicates of 0.35 μM (10.9 $\mu\text{g P/L}$) o-phosphate standard in synthetic seawater***

Concentration (mg P/L)				
	0.365	0.318	0.395	0.325
	0.363	0.324	0.365	0.315
	0.363	0.306	0.363	0.324
	0.372	0.305	0.363	0.328
	0.364	0.324	0.372	0.325
	0.373	0.321	0.364	0.327
	0.369	0.319	0.373	0.331
	0.372	0.313	0.369	0.347
	0.378	0.307	0.372	0.336
	0.389	0.323	0.378	0.336
Statistics				
Mean	0.371	0.316	0.371	0.330
Std. Deviation	0.008	0.007	0.010	0.009
% RSD**	2.2	2.4	2.6	2.6
% Recovery	105.9	90.3	106.1	94.1

14.3 PRECISION STUDY

Replicates of 3.5 μM (108 $\mu\text{g P/L}$) o-phosphate standard in synthetic seawater ***

Concentration (mg P/L)				
Statistics				
Mean	3.470	3.389	3.383	3.334
% Recovery	99.1	96.8	96.7	95.3
Std. Deviation	0.079	0.022	0.014	0.028
% RSD**	2.3	0.6	0.4	0.8
Maximum	3.627	3.418	3.409	3.376
Minimum	3.309	3.345	3.357	3.304

Replicates of 7.0 $\mu\text{M P}$ (216 $\mu\text{g P/L}$) o-phosphate standard in synthetic seawater ***

Concentration (mg P/L)				
Statistics				
Mean	6.911	6.889	6.714	6.980
% Recovery	98.7	98.4	95.9	99.7
Std. Deviation	0.033	0.039	0.028	0.047
% RSD**	0.5	0.6	0.4	0.7
Maximum	6.970	6.948	6.764	7.075
Minimum	6.873	6.825	6.684	6.888

* MDL is calculated by multiplying the Standard Deviation by the Student-t value for the number of replicates run.

** %RSD is calculated by dividing the Standard Deviation by the mean, then multiplying by 100.

*** To prepare these standards, refer to Section 9.

15 REVISIONS

December 2019	Rev. 0	Method created and support data acquired
April 2020	Rev. 1	Minor editorial changes Added sensitivity and blank absorbance information